

Around the Boundary of Science*

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Translated from the Danish

The negative result of *Michelson's* attempt to demonstrate the earth's translation by the aid of the displacement of the interference fringes led *Einstein* — in 1905 — to set up his "*Principle of Relativity*"**):

The velocity with which electromagnetic vibrations propagate themselves in space is observed as *the same* by two observers, of whom the one has a uniform motion in a straight line in relation to the other — a presupposition which was added to the usual one: the vibrations are propagated with *the same* velocity in all directions, independently of the source's motion.

But this assertion of Einstein's has divided the physicists into two camps, which are constantly at strife.

The opponents maintain that the two presuppositions stand in mutual contradiction, and that the assertion is therefore false, while the adherents seek to show that the contradiction is only apparent, and therefore claim that the assertion be accepted as a scientific hypothesis.

The purpose of the present article is to recall that it is not excluded *in advance* that both parties are right in the *essential* matter. It is possible that the opponents are right that the two presuppositions really are contradictory (without the assertion therefore needing to be false), and that the adherents are right that the assertion is a well-founded hypothesis (without this therefore needing to be scientific).

It must not be forgotten that we dare not assert that existence is "comprehensible or intelligible, measurable for our thought" (Høffding). We cannot know in advance

*The Editors have not wished to refuse this article admission, although it must be said to fall outside the journal's frame, but reserve their position with regard to any further contributions on the questions raised in it.

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whether there may not — in spite of everything — be found contradictions in the world of things.

The condition for our being able to apply logic to reality is that the latter, at all events within *the domain in question*, contains no contradictions. But it is still only a *hypothesis* that in *the whole* of existence, taken as one, no contradictions are to be found — a hypothesis which, moreover, seems difficult to maintain together with the hypothesis of the world-ether.

When we live in a world where the state of things is insanely improbable (Bjerrum), it is hardly quite improbable that existence taken as one really is illogical, irrational, however painful it might be for us to acknowledge the possibility that what is irreconcilable for human consciousness is nevertheless found reconciled in existence, and conversely. But our knowledge of the real may all the same very well be *piecewise* entirely correct.

The atomic, electron, quantum and mutation theories, together with the sudden new formations which can arise in the psychic domain, teach an indivisibility and discontinuity in existence which for thought is quite enigmatic. It is possible that our knowledge of the real — like reality itself — will show itself “discontinuous”, that we can know existence only *by leaps*, that we must *leap* from the one domain to the other, because there is no point of transition. One of the consequences of the principle of relativity is that two observers can under certain conditions judge contradictorily about the same phenomenon and yet both be right, each from his own point of view¹. It is possible that a phenomenon, particularly one from the border region between two domains of knowledge, is judged contradictorily in a similar way from the respective points of view, and that there is for the two domains no *common* point from which the phenomenon can be seen; but no one can at the *same* time occupy both points of view, just as no one can simultaneously be both at rest and in motion.

Even if existence should thus be knowable only piecewise, it does not follow from this that such “contradictions” are not reconcilable for an intelligence of a higher and different kind, nor does it on the other hand mean that there is a limit to how far science can press forward in piecewise knowledge (even though large domains are probably not knowable by us at all).

By its very nature science can never give up before any task, not even if it should meet with one of which it would never be able to reach any solution, unless the insolubility were scientifically proved.

But perhaps it is such an insoluble task to reconcile the two presuppositions of the

¹Vol. 10, p. 266.

principle of relativity. Perhaps the electromagnetic vibrations are phenomena on the boundary between two different domains, between sensible nature and the insensible ether.

The presupposition (hypothesis) of the constancy of the velocity has in that case been arrived at with the ether as starting point, while the starting point for the other presupposition (the relativity hypothesis itself) is in sensible nature. (As *Wiechert* has shown, the principle of relativity does not exclude the ether hypothesis; on the contrary, the principle compels its maintenance).

Thus: it is possible that the two presuppositions really are contradictory *without* the one of them therefore needing to be false.

It seems sufficiently established that the hypothesis is illogical and entails a “causeless phenomenon”^{**}), but thereby it is condemned only as a *scientific* hypothesis, “whereas surely every “thinking being” will perceive that because the causeless is unknowable, it is not therefore given that there is nothing causeless” (Kroman)..

Science is only one of the servants given to man as a help towards living life. “Science and philosophy are not enough for a personal view of life” (Høffding). “By a purely individual route too, through feeling and personal experience of life, we obtain conviction of the truth of many an assertion” (Kroman). The scientific mode of knowledge moreover gives us only a *relative*, no absolute knowledge, since it consists in an analysis whose result depends on the point of view chosen and the symbols used (Bergson). But in the other modes of knowledge too, use is made of the principle of causality and of logical thinking *within* the domains in question; only the observations are often of another nature than science’s, just as the starting point does not always lie in the objective alone. And science as such cannot occupy the points of view in question; they lie *out side* its boundary.

Since man is after all only *partially* identical with the man of science, man as such cannot *content himself* with taking science’s point of view. There is thus a possibility that it is *the man* Einstein, and not the man of science in him, who has intuitively discerned a hint in science’s results and, led by it, has unconsciously set up the relativity hypothesis outside science’s boundary. And Michelson’s result, as well as the dependence of the electron mass upon the velocity in the β -rays, is undeniably weighty circumstantial evidence for its correctness.

Thus: it is possible that the hypothesis really is well founded *without* its therefore needing to be a *scientific* hypothesis.

^{**})Vol. 14, p 17 and 28.

Should it really be the case that we have here an unscientific, illogical hypothesis with a probability of being nevertheless true and valid for certain phenomena of nature, then the relativists commit the error of attempting the impossible: to press the hypothesis within science's frame, while the opponents for their part make themselves guilty of the error of declaring the hypothesis false. And confusion arises, with incessant misunderstandings and idle strife, when the parties do not perceive this and concede to each other that the hypothesis is illogical, and respectively that it may nevertheless be true.

But it is hardly to be expected that this intimation, from a third party, of a possibility of relieving the tension and settling the strife between the two parties by such a consideration will find understanding with either of them. Perhaps the possibility will be disputed by the objection that it is meaningless to speak of logical consequences of an illogical hypothesis. Science cannot operate with such a thing at all.

This objection would however be premature, for just as it is science's own results that have led to the hypothesis, so it will also be possible to test it and its consequences upon *other* of science's results, and to seek further circumstantial evidence for its validity.

And if new observations are successfully made which agree with it, this presumably indicates that the phenomena in question also belong to the border region between sensible nature and the insensible ether.

Outside natural science too, for phenomena on the boundary between the subjective and the objective, the principle of relativity has validity. Thus it holds for the problem of the "freedom" of the will. Starting from the objective, one comes with logical necessity to determinism; but starting from the subjective, one comes just as unavoidably to indeterminism; in other words, the phenomenon of will is judged contradictorily by an objective and by a subjective consideration.

Determinism is right, but indeterminism is right too. This man as such can concede; but the man of science is inconsistent if he — as some do — regards the problem as scientifically insoluble, for objective science does not build upon subjective experiences.

The problem has *two* solutions, but science can, by its presupposition, arrive only at the *one*: determinism.

In a similar way the principle of relativity holds for the relation between the scientific view of the world and the intuitive one (Bergson).

In spite of real contradictions, both views may be correct.

And the old strife about the religious problem can in part be explained in the same way.